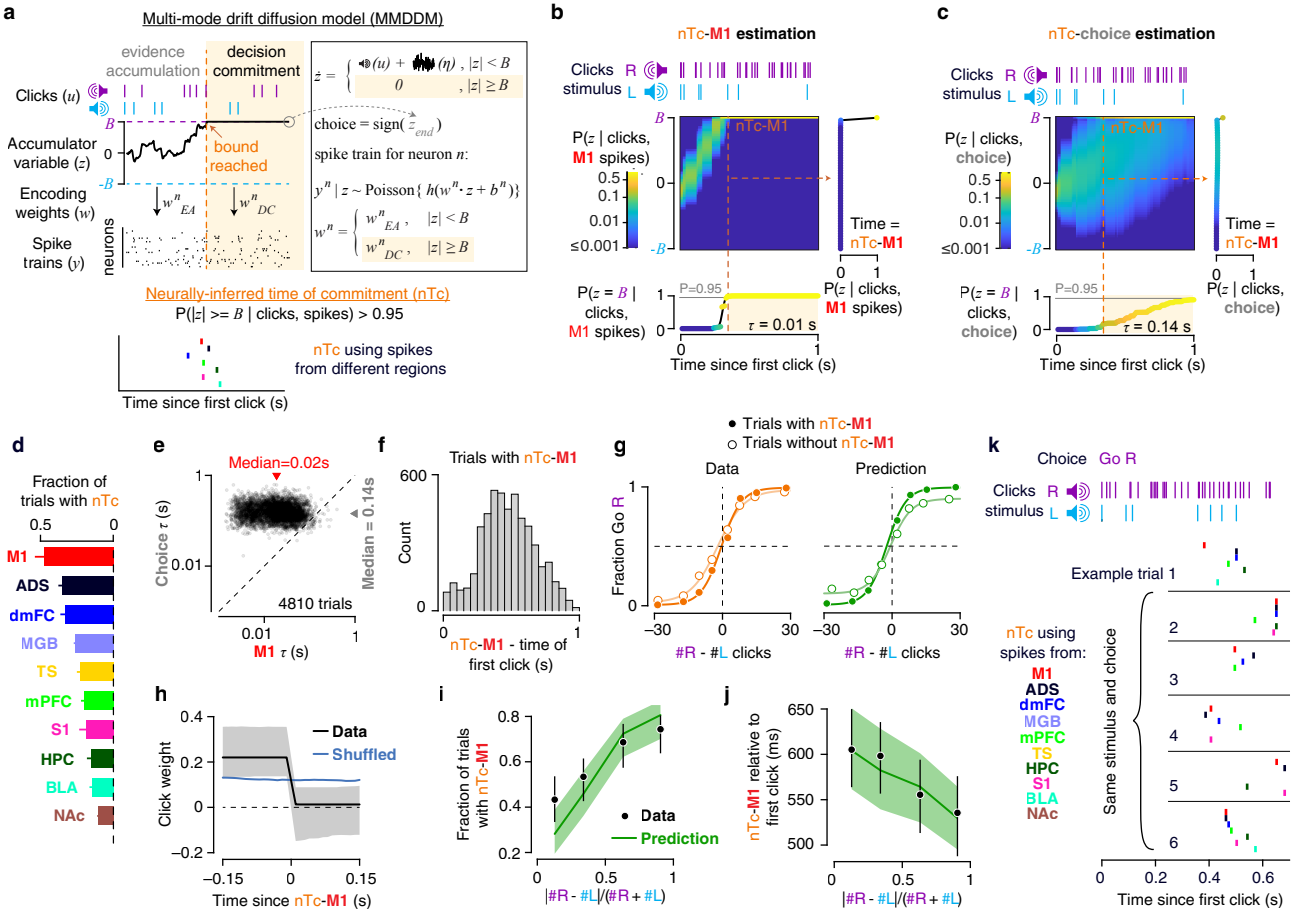
a
Multi-mode drift diffusion model (MMDDM)
evidence accumulation
decision commitment
Clicks (u)
Accumulator variable (z)
Encoding weights (w)
Spike trains (y)
neurons
bound reached
choice = sign(z_end)
spike train for neuron n:
Neurally-inferred time of commitment (nTc)
P(|z| >= B | clicks, spikes) > 0.95
nTc using spikes from different regions
Time since first click (s)
b
nTc-M1 estimation
Clicks stimulus
P(z | clicks, M1 spikes)
nTc-M1
Time = nTc-M1
P(z = B | clicks, M1 spikes)
P=0.95
τ = 0.01 s
P(z | clicks, M1 spikes)
Time since first click (s)
c
nTc-choice estimation
Clicks stimulus
P(z | clicks, choice)
nTc-M1
Time = nTc-M1
P(z = B | clicks, choice)
P=0.95
τ = 0.14 s
P(z | clicks, choice)
Time since first click (s)
d
Fraction of trials with nTc
M1
ADS
dmFC
MGB
TS
mPFC
S1
HPC
BLA
NAc
e
Median=0.02s
Median = 0.14s
Choice τ (s)
4810 trials
M1 τ (s)
f
Trials with nTc-M1
Count
nTc-M1 - time of first click (s)
g
Trials with nTc-M1
Trials without nTc-M1
Data
Prediction
Fraction Go R
#R - #L clicks
h
Data
Shuffled
Click weight
Time since nTc-M1 (s)
i
Fraction of trials with nTc-M1
Data
Prediction
|#R - #L|/(#R + #L)
j
nTc-M1 relative to first click (ms)
|#R - #L|/(#R + #L)
k
Choice Go R
Clicks stimulus
Example trial 1
nTc using spikes from:
M1
ADS
dmFC
MGB
mPFC
TS
HPC
S1
BLA
NAc
Same stimulus and choice
Time since first click (s)